

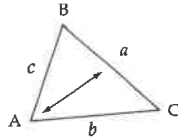
The sine rule: mixed questions

6.1

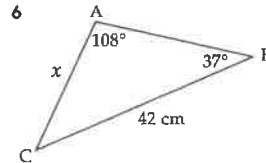
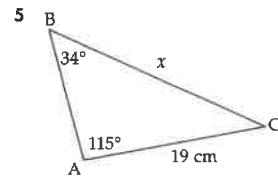
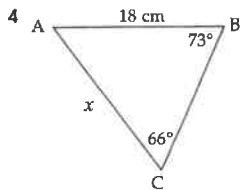
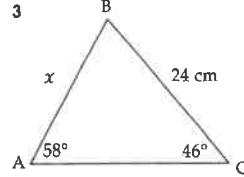
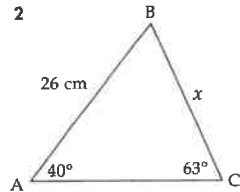
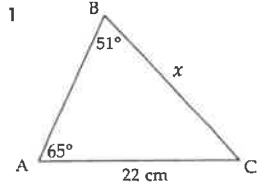
A

The sine rule states that in any triangle ABC:

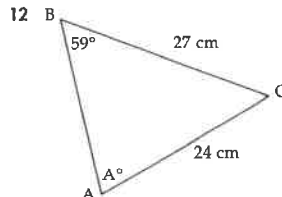
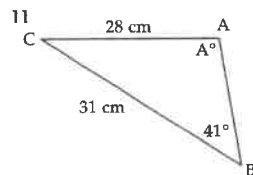
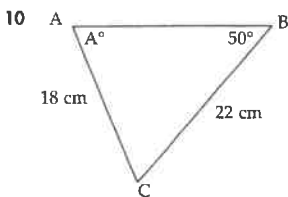
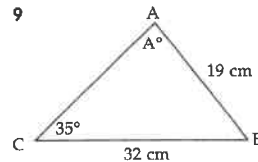
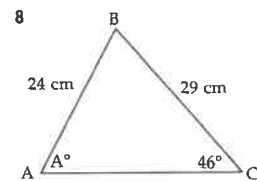
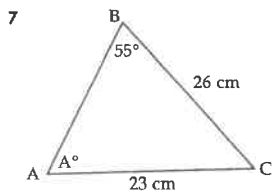
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$



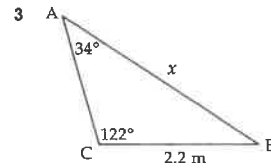
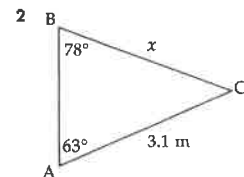
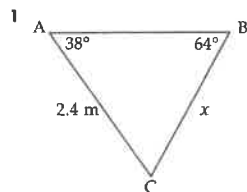
Use the sine rule to calculate the length of side x correct to 1 decimal place.



Use the sine rule to calculate the size of angle A to the nearest degree.

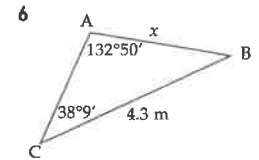
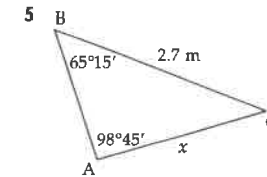
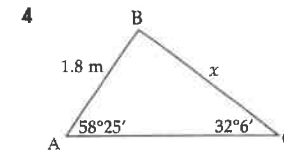


B Use the sine rule to calculate the length of side x correct to 1 decimal place.

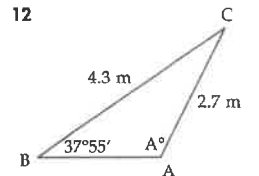
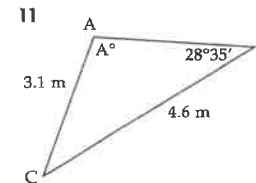
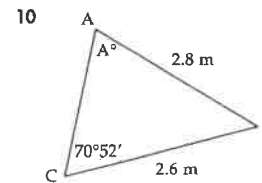
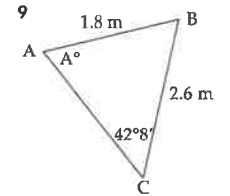
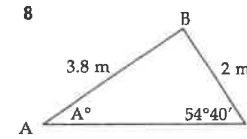
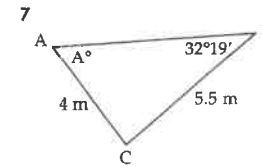


6.2

Use the sine rule to calculate the length of side x correct to 1 decimal place.



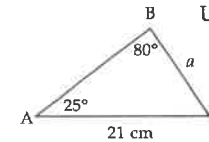
Use the sine rule to find the size of angle A in degrees and minutes.



C

EXAMPLE:

In the triangle ABC, $A = 25^\circ$,
 $B = 80^\circ$ and $b = 21$ cm.
Find the length of side a
to 1 decimal place.



Using the sine rule:

$$\frac{a}{\sin 25^\circ} = \frac{21}{\sin 80^\circ}$$

rearrange: $a = \frac{21 \times \sin 25^\circ}{\sin 80^\circ}$
 $= 9.0$ cm

For each question, draw a triangle ABC and label it with the given information. Then calculate the length of side a correct to 1 decimal place.

1 $A = 48^\circ$, $B = 72^\circ$ and $b = 15$ cm

2 $A = 62^\circ$, $B = 53^\circ$ and $b = 17$ cm

3 $A = 112^\circ$, $B = 31^\circ$ and $b = 13$ cm

4 $A = 121^\circ$, $C = 40^\circ$ and $c = 23$ cm

Find the size of angle A in degrees and minutes. (A is an acute angle.)

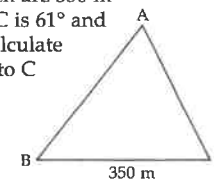
5 $B = 35^\circ$, $a = 15$ cm and $b = 13$ cm

6 $B = 61^\circ$, $a = 47$ cm and $b = 44$ cm

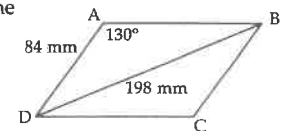
7 $B = 92^\circ$, $a = 32$ cm and $b = 48$ cm

8 $C = 42^\circ$, $a = 39$ cm and $c = 31$ cm

9 A steeple (A) is observed from the two points B and C, which are 350 m apart. The angle ABC is 61° and angle ACB is 73° . Calculate the distance from A to C to the nearest metre.



10 In the parallelogram ABCD, $AD = 84$ mm, angle DAB = 130° and the diagonal BD = 198 mm. Calculate the size of angle ABD to the nearest degree.



The cosine rule: mixed questions

9.1

A

The **cosine rule** states that in any triangle ABC:

$$a^2 = b^2 + c^2 - 2bc \cos A$$

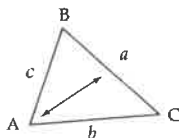
$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

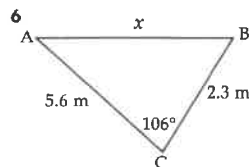
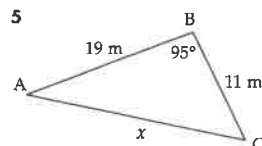
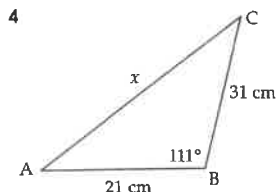
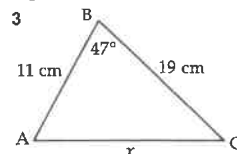
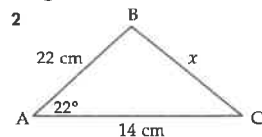
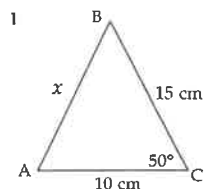
$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

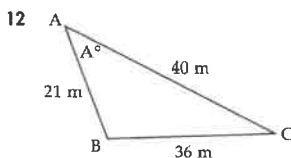
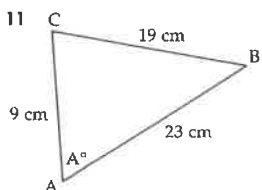
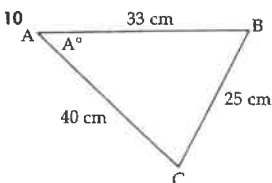
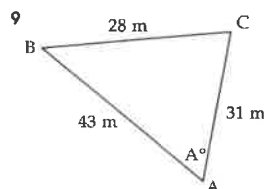
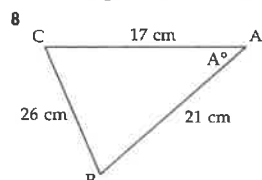
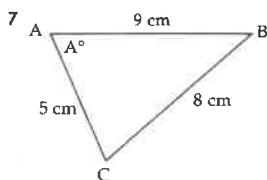
$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$



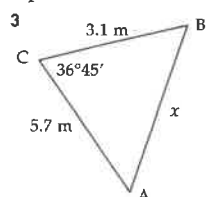
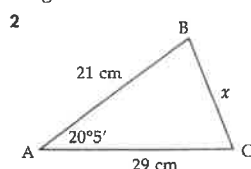
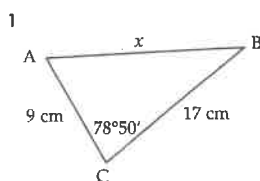
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Use the cosine rule to calculate the size of angle A to the nearest degree.

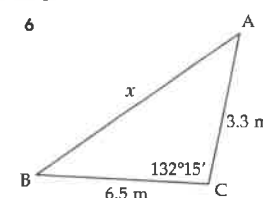
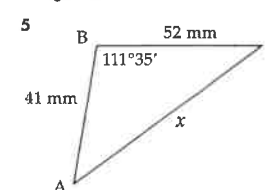
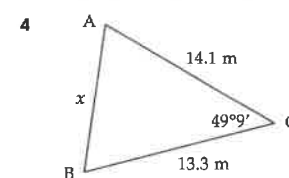


B Use the cosine rule to calculate the length of side x correct to 1 decimal place.

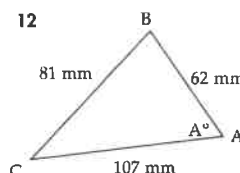
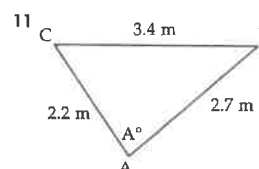
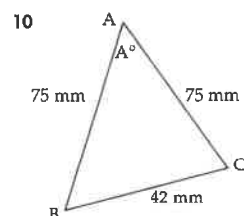
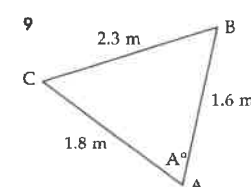
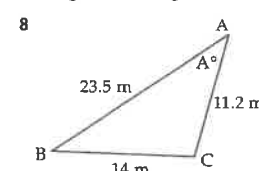
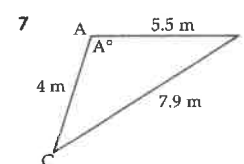


9.2

Use the cosine rule to calculate the length of side x correct to 1 decimal place.



Use the cosine rule to find the size of angle A in degrees and minutes.

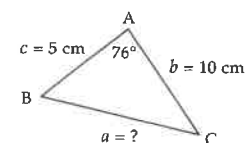


C

EXAMPLE:

In the triangle ABC, $A = 76^\circ$, $b = 10$ cm and $c = 5$ cm. Calculate the length of side a correct to 1 decimal place.

Use the cosine rule because you are given the lengths of 2 sides and the included angle. Draw and label a sketch.



$$\begin{aligned} a^2 &= b^2 + c^2 - 2bc \cos A \\ &= 10^2 + 5^2 - 2(10)(5) \cos 76^\circ \\ a &= 10.0 \text{ cm} \end{aligned}$$

For each question, draw a triangle ABC and label it with the given information. Then calculate the length of side a correct to 1 decimal place.

1 $A = 56^\circ$, $b = 6$ cm and $c = 11$ cm

2 $A = 28^\circ$, $b = 33$ cm and $c = 41$ cm

3 $A = 23^\circ 35'$, $b = 19.2$ m and $c = 11.4$ m

4 $A = 77^\circ 15'$, $b = 23.4$ m and $c = 37.6$ m

Find the size of angle A in degrees and minutes.

5 $a = 15$ cm, $b = 13$ cm and $c = 18$ cm

6 $a = 112$ cm, $b = 32$ cm and $c = 90$ cm

7 $b = 14.1$ m, $c = 9.7$ m and $a = 19.6$ m

8 $c = 1.6$ m, $a = 3.3$ m and $b = 3.5$ m